問題 262

次の問いに答えよ．

解答:

$\sin^2\alpha+\cos^2\alpha=1$ で残りを求める．

(1) α 鋭角，$\sin\alpha=\dfrac{1}{4}$：$\cos\alpha=\dfrac{\sqrt{15}}{4}$，$\tan\alpha=\dfrac{1}{\sqrt{15}}=\dfrac{\sqrt{15}}{15}$

(2) α 鈍角, $\sin\alpha=\dfrac{1}{4}$：$\cos\alpha=-\dfrac{\sqrt{15}}{4}$, $\tan\alpha=-\dfrac{\sqrt{15}}{15}$

(3) α 鈍角, $\cos\alpha=-\dfrac{5}{6}$：$\sin\alpha=\dfrac{\sqrt{11}}{6}$, $\tan\alpha=-\dfrac{\sqrt{11}}{5}$

問題 263

$\tan\alpha$ から $\sin\alpha$，$\cos\alpha$ を求めよ．

解答:

$1+\tan^2\alpha=\dfrac{1}{\cos^2\alpha}$ を使う．

(1) α 鋭角，$\tan\alpha=\dfrac{1}{3}$

$\cos^2\alpha=\dfrac{9}{10}$，$\cos\alpha=\dfrac{3\sqrt{10}}{10}$，$\sin\alpha=\dfrac{\sqrt{10}}{10}$

(2) α 鈍角，$\tan\alpha=-2$

$\cos^2\alpha=\dfrac{1}{5}$，$\cos\alpha=-\dfrac{\sqrt{5}}{5}$，$\sin\alpha=\dfrac{2\sqrt{5}}{5}$

問題 264

△ABC において，正弦定理を用いて求めよ．

解答:

$\dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}=2R$

(1) $b=4$，$A=30^\circ$，$B=45^\circ$

$\dfrac{a}{\sin 30^\circ}=\dfrac{4}{\sin 45^\circ}$，$a=\dfrac{4\sin 30^\circ}{\sin 45^\circ}=\dfrac{2}{\frac{\sqrt{2}}{2}}=2\sqrt{2}$

(2) $b=2$，$c=\sqrt{3}$，$B=45^\circ$

$\dfrac{2}{\sin 45^\circ}=\dfrac{\sqrt{3}}{\sin C}$，$\sin C=\dfrac{\sqrt{3}\cdot\frac{\sqrt{2}}{2}}{2}=\dfrac{\sqrt{6}}{4}$

(3) $c=5$，$A=45^\circ$，$B=105^\circ$

$C=30^\circ$. $\dfrac{a}{\sin 45^\circ}=\dfrac{5}{\sin 30^\circ}=10$, $a=10\cdot\dfrac{\sqrt{2}}{2}=5\sqrt{2}$

問題 265

1 辺が a の正三角形の外接円の半径を求めよ．

解答:

$2R=\dfrac{a}{\sin 60^\circ}=\dfrac{2a}{\sqrt{3}}\to R=\dfrac{\sqrt{3}\,a}{3}$

問題 266

余弦定理を用いて辺の長さを求めよ．

解答:

$a^2=b^2+c^2-2bc\cos A$

(1) $b=4$，$c=\sqrt{3}$，$A=30^\circ$

$a^2=16+3-8\sqrt{3}\cdot\dfrac{\sqrt{3}}{2}=19-12=7$，$a=\sqrt{7}$

(2) $a=\sqrt{3}$，$c=\sqrt{6}$，$B=135^\circ$

$b^2=3+6-2\sqrt{18}\cdot\left(-\dfrac{\sqrt{2}}{2}\right)=9+6=15$，$b=\sqrt{15}$

(3) $a=3\sqrt{3}$，$b=2$，$C=150^\circ$

$c^2=27+4-12\sqrt{3}\cdot\left(-\dfrac{\sqrt{3}}{2}\right)=31+18=49$，$c=7$

問題 267

$a=2$，$b=4$，$c=5$ のとき $\cos A$，$\cos B$，$\cos C$ を求めよ．

解答:

$\cos A=\dfrac{b^2+c^2-a^2}{2bc}=\dfrac{16+25-4}{40}=\dfrac{37}{40}$

$\cos B=\dfrac{a^2+c^2-b^2}{2ac}=\dfrac{4+25-16}{20}=\dfrac{13}{20}$

$\cos C=\dfrac{a^2+b^2-c^2}{2ab}=\dfrac{4+16-25}{16}=-\dfrac{5}{16}$

問題 268

△ABC の面積を求めよ．

解答:

$S=\dfrac{1}{2}bc\sin A$

(1) $b=5$，$c=7$，$A=45^\circ$：$S=\dfrac{35}{2}\cdot\dfrac{\sqrt{2}}{2}=\dfrac{35\sqrt{2}}{4}$

(2) $a=2$，$b=3$，$C=150^\circ$：$S=\dfrac{1}{2}\cdot 2\cdot 3\cdot\dfrac{1}{2}=\dfrac{3}{2}$

問題 269

$a=7$，$B=30^\circ$，面積 9 のとき c を求めよ．

解答:

$9=\dfrac{1}{2}\cdot 7\cdot c\cdot\sin 30^\circ=\dfrac{7c}{4}\to c=\dfrac{36}{7}$

問題 270

$a=5$，$b=6$，$c=9$ のとき．

解答:

(1) $\cos C=\dfrac{25+36-81}{60}=-\dfrac{1}{3}$

(2) $\sin C=\sqrt{1-\dfrac{1}{9}}=\dfrac{2\sqrt{2}}{3}$

(3) $S=\dfrac{1}{2}\cdot 5\cdot 6\cdot\dfrac{2\sqrt{2}}{3}=10\sqrt{2}$

問題 271

ヘロンの公式で面積を求めよ.

解答:

$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \dfrac{a+b+c}{2}$

(1) $a = 5, \quad b = 7, \quad c = 8 : s = 10$

$S = \sqrt{10 \cdot 5 \cdot 3 \cdot 2} = \sqrt{300} = 10\sqrt{3}$

(2) $a = 2, \quad b = 3, \quad c = 4 : s = \dfrac{9}{2}$

$S = \sqrt{\dfrac{9}{2} \cdot \dfrac{5}{2} \cdot \dfrac{3}{2} \cdot \dfrac{1}{2}} = \dfrac{3\sqrt{15}}{4}$